

## Data Engineering, Business, and Systems Theory

Data Engineering plays a crucial role in today's business landscape, facilitating the collection, storage, and processing of vast amounts of data to generate meaningful insights. However, its impact extends beyond technical processes. By understanding the broader organizational ecosystem and systems theory, data engineers can design infrastructure that not only meets technical needs but also aligns with business objectives.

In this lesson, we will explore how data engineering fits within business processes and systems theory, and reflect on its value to organizations.

### Learning Objectives

By the end of this train, you should be able to:

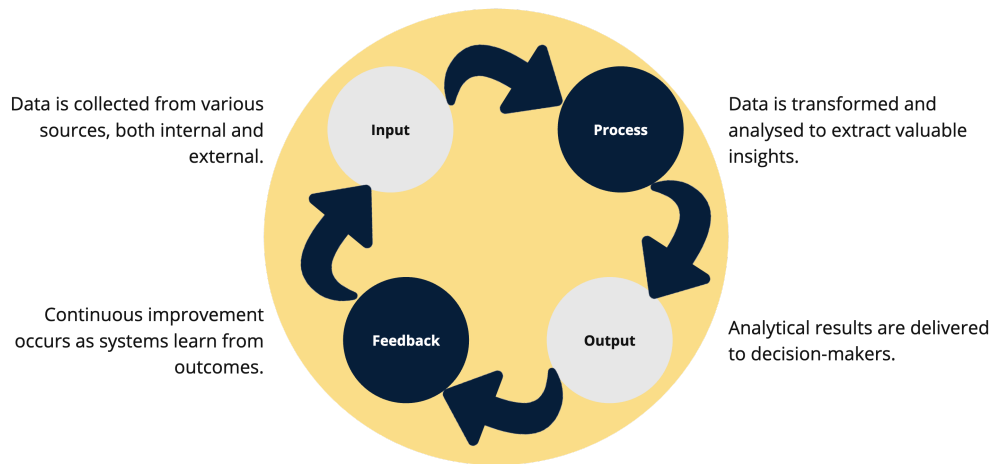
- Understand the different types of organizational systems and their functions.
- Explain the flow of data through these systems and its impact on decision-making.
- Describe the relationship between data engineering, business objectives, and systems theory.
- Recognize the variety of systems within different types of industries.

### Introduction to Organizational Systems

Every business operates as a system, where different departments, teams, and technologies interact to achieve common goals. In systems theory, organizations are viewed as complex, interconnected entities. This holistic perspective encourages us to focus on how data engineering contributes to the overall success of a company by enabling better decision-making, improving efficiency, and enhancing customer experiences.

### Systems Theory in Business

Systems theory says that an organization functions like an ecosystem — composed of subsystems (departments, resources, and technologies) that must work together. Each subsystem is interdependent, and changes in one part of the system affect the others. Data engineering is a subsystem that plays a vital role in ensuring smooth data flow across departments, thereby supporting the broader organizational system.



## Types of Organisational Systems

Organizations use a variety of systems to manage and analyze data. These systems support different functions, from day-to-day operations to high-level strategic planning. Below are the main categories of organisational systems:

### Transaction Processing Systems (TPS):

TPS are systems that manage the everyday activities of a business, like handling sales transactions, processing payroll, and managing shipments. Think of them as the systems that keep the business running smoothly day-to-day.

### Management Information Systems (MIS):

MIS provides managers with regular reports on the company's performance, helping them make informed decisions. These reports might show trends in sales, inventory levels, or employee productivity, giving a snapshot of how the business is doing.

### Enterprise Resource Planning (ERP):

ERPs integrate information from across the company – from finance to manufacturing to HR. This way, everyone in the organization is on the same page, working with the same data. It's like a central hub that keeps everything connected and coordinated.

### Customer Relationship Management (CRM):

CRMs store and organise information about customers, such as past purchases and interactions. By having customer details in one place, businesses can personalize service, address customer needs better, and strengthen relationships.

## Data Flow Through Organizational Systems

Data flows through these systems in different ways, depending on the architecture. For example, an organization might collect transactional data from an ERP, process it in a data warehouse for reporting, and feed it into a CRM for enhanced customer targeting. Each

system plays a role in collecting, storing, and moving data to where it can be best utilized for decision-making.

## The Role of Data Engineering in Organizational Systems

Data engineers design and maintain data pipelines and databases that enable seamless data exchange between subsystems, ensuring data is collected, stored, processed, and made available to decision-makers in real-time or near real-time. By ensuring that data is reliable and accessible, data engineers empower businesses to make informed decisions that drive growth and competitive advantage.

### Reflection

In what ways do you think a data engineer could enhance the functionality of these systems to support business objectives?

## Data Engineers as Problem Solvers within Business Systems

One of the key tenets of systems theory is that subsystems must continuously adapt and evolve to keep the overall system functional. Data engineers often act as problem solvers, ensuring that data infrastructure adapts to the changing needs of the business. For instance, as a company scales, its data volume may increase exponentially, requiring engineers to optimize pipelines, upgrade storage solutions, or implement distributed computing systems. This adaptability is not only technical but also strategic. Data engineers must understand the business's goals to prioritize projects and allocate resources effectively.

### For example:

When a business decides to expand into a new market, data engineers may be tasked with ensuring that the data infrastructure can handle new customer data from that region. Data engineers might need to strategically enhance data pipelines to accommodate increased customer data while aligning their efforts with the organization's goals. This involves planning for scalability, integrating new systems, and ensuring compliance with local regulations.

## Organizational Systems for Data Management and Analysis

Different types of organisations use a wide range of systems to manage and analyze data, based on their industry-specific needs. Here are a few systems used in the healthcare, telecommunications and utilities industries:

## Healthcare

**Systems:** In the healthcare industry, organizations employ systems such as Electronic Health Records (EHR), clinical decision support tools, and healthcare customer relationship management (CRM) systems.

**Data flow:** Patient information is collected through EHR systems, which allows healthcare providers to analyse this data to improve health outcomes and reduce costs. By leveraging insights from patient data, hospitals and clinics can enhance their services and deliver more effective care tailored to individual needs.

## Telecommunications

**Systems:** The telecommunications sector relies on customer data platforms and network management systems to optimise its operations.

**Data flow:** Telecom companies process vast amounts of customer and network data, allowing them to analyze usage patterns and service performance. This data-driven approach enables these companies to improve service quality, make informed decisions about network enhancements, and reduce customer churn by addressing service issues proactively.

## Utilities

**Systems:** In the utilities sector, organizations use smart grid technology and asset management platforms to efficiently monitor and manage resources.

**Data flow:** Data collected from smart meters and energy management systems is analysed to optimise energy distribution and predict when maintenance is needed. This proactive data analysis helps utilities ensure a reliable energy supply and reduce downtime, ultimately benefiting customers with improved service.

Every time you binge-watch a series on Netflix, sophisticated data engineering systems are at work behind the scenes, analysing millions of viewing preferences. Netflix uses these insights not only to recommend your next favourite show but also to decide which new shows to produce, resulting in tailored content for over 200 million users worldwide!

### Case Study: Netflix and data-driven decision making

This case study explores how Netflix uses data engineering and analytics to drive its business strategy. Netflix collects vast amounts of data on user preferences, viewing habits, and engagement metrics. By leveraging this data, Netflix makes informed decisions about content creation, marketing strategies, and user experience improvements.

### Video: How Netflix Uses Big Data

Video: YouTube video player

[https://www.youtube.com/embed/jjn\\_hv\\_wxgc?si=f0vDbVn6nj0cj6ls](https://www.youtube.com/embed/jjn_hv_wxgc?si=f0vDbVn6nj0cj6ls)

## Conclusion

Understanding the synergy between **data engineering**, **business operations**, and **systems theory** provides valuable insights into how organizations can harness data to create value. Data engineers are crucial to enabling efficient, data-driven processes that drive business success.

### Key Takeaways

- Takeaway 1: Data engineers make systems that move and prepare data for use.
- Takeaway 2: They make sure data is safe, good quality, and easy to work with.
- Takeaway 3: Big companies like Amazon and HSBC succeed because of good data work.
- Takeaway 4: Data engineering makes raw information useful for any industry.
- Takeaway 5: Good data helps people make better choices in business.